

Western Diet Consumption and Cognitive Impairment: Links to Hippocampal Dysfunction and Obesity

Intake of saturated fats and simple carbohydrates, two of the primary components of a modern Western diet, is linked with the development of obesity and Alzheimer's Disease. The present paper summarizes research showing that Western diet intake is associated with cognitive impairment, with a specific emphasis on learning and memory functions that are dependent on the integrity of the hippocampus. The paper then considers evidence that saturated fat and simple carbohydrate intake is correlated with neurobiological changes in the hippocampus that may be related to the ability of these dietary components to impair cognitive function. Finally, a model is described proposing that Western diet consumption contributes to the development of excessive food intake and obesity, in part, by interfering with a type of hippocampal-dependent memory inhibition that is critical in the ability of animals to refrain from responding to environmental cues associated with food, and ultimately from consuming energy intake in excess of that driven solely by caloric need.